Tutorial 2 & 3

Create and declare the variables to capture the input from users.

1. Write a C++ program to find and print the Area and Perimeter of a Rectangle. Prompt the user to enter length and width of the rectangle and calculate using the formula given below.
   area = length * width
   perimeter = 2(l+w)

2. Write a C++ program to convert the temperature in Celsius to Fahrenheit. Prompt the user to enter a temperature in Celsius and convert it using the following formula F = 9/5*C + 32, where C is temperature in Celsius and F is temperature in Fahrenheit.

3. Write a C++ program to prompt the user to input of an integer value greater than 1000. Then multiply the number by itself and display the result.

4. Write a program in C++ that converts kilometers to miles. Prompt the user to enter kilometers and display in miles. Note that one kilometer is 0.621371 miles.

5. Write a C++ program to find the third angle of a triangle. Create variables to capture 1st and 2nd angle entered by user. (The sum of the interior angles of any triangle in Euclidean space is always 180)

   Sample Output:

   ```
   Find the third angle of a triangle:
   ----------------------------------------
   Input the 1st angle of the triangle: 30
   Input the 2nd angle of the triangle: 60
   The 3rd angle of this triangle is: 90
   ```

6. (Health application: BMI) Body Mass Index (BMI) is a measure of health on weight. Write a C++ program that prompts the user to enter a weight in pounds and height in inches and displays the BMI. Note that one pound is 0.4536 kilograms and one inch is 0.0254 meters.

BMI = kg/m²

(use header file #include<math.h> and pow(a, b) to define a^b)

sample output:
Enter weight in pounds: 95.5
Enter height in inches: 50
BMI is 26.8577